

Practice Problems on MLP and Backpropagation

CSE427: Machine Learning

Instructions

Solve all problems step by step. Show intermediate calculations clearly.

Topics covered:

- MLP forward pass
- ReLU activation
- Squared-error loss
- Output-layer gradients
- Hidden-layer gradients
- Backpropagation using chain rule
- Parameter update using gradient descent
- Softmax and cross-entropy
- Multi-class output interpretation

Common Network Structure

For all problems, consider the following MLP unless otherwise stated.

There are two input nodes:

$$X_1, \quad X_2$$

There are two hidden nodes:

$$Y_3, \quad Y_4$$

There are three output nodes:

$$Y_5, \quad Y_6, \quad Y_7$$

Hidden layer equations:

$$z_3 = W_{13}X_1 + W_{23}X_2 + b_3$$

$$Y_3 = \text{ReLU}(z_3)$$

$$z_4 = W_{14}X_1 + W_{24}X_2 + b_4$$

$$Y_4 = \text{ReLU}(z_4)$$

Output layer pre-activation equations:

$$z_5 = W_{35}Y_3 + W_{45}Y_4 + b_5$$

$$z_6 = W_{36}Y_3 + W_{46}Y_4 + b_6$$

$$z_7 = W_{37}Y_3 + W_{47}Y_4 + b_7$$

ReLU function:

$$\text{ReLU}(z) = \begin{cases} z, & z > 0 \\ 0, & z \leq 0 \end{cases}$$

Gradient descent update rule:

$$\theta^{new} = \theta^{old} - \eta \frac{\partial Loss}{\partial \theta}$$

Part A: Forward Pass and ReLU

Practice Problem

Problem 1: Basic Hidden Layer Forward Pass

Given:

$$X_1 = 2, \quad X_2 = 1$$

$$W_{13} = 0.5, \quad W_{23} = -0.4, \quad b_3 = 0.2$$

$$W_{14} = -0.3, \quad W_{24} = 0.8, \quad b_4 = -0.1$$

Calculate:

$$z_3, \quad Y_3, \quad z_4, \quad Y_4$$

Also state which hidden node is active after ReLU.

Practice Problem

Problem 2: ReLU Blocking Case

Given:

$$X_1 = 1, \quad X_2 = 3$$

$$W_{13} = -1, \quad W_{23} = 0.2, \quad b_3 = 0.1$$

$$W_{14} = 0.4, \quad W_{24} = -0.6, \quad b_4 = 1$$

Calculate:

$$z_3, \quad Y_3, \quad z_4, \quad Y_4$$

Then answer:

- (a) Which hidden node becomes zero?
- (b) What will be the value of $\text{ReLU}'(z)$ for that node?
- (c) Why is this important for backpropagation?

Practice Problem

Problem 3: Full Forward Pass with ReLU Output

Given:

$$X_1 = 1, \quad X_2 = 2$$

Hidden layer:

$$W_{13} = 0.6, \quad W_{23} = 0.2, \quad b_3 = -0.1$$

$$W_{14} = -0.5, \quad W_{24} = 0.7, \quad b_4 = 0.3$$

Output layer:

$$W_{35} = 0.4, \quad W_{45} = -0.2, \quad b_5 = 0.1$$

$$W_{36} = -0.3, \quad W_{46} = 0.5, \quad b_6 = -0.2$$

$$W_{37} = 0.2, \quad W_{47} = 0.6, \quad b_7 = 0.0$$

The output layer also uses ReLU:

$$Y_5 = \text{ReLU}(z_5)$$

$$Y_6 = \text{ReLU}(z_6)$$

$$Y_7 = \text{ReLU}(z_7)$$

Calculate:

$$Y_3, \quad Y_4, \quad Y_5, \quad Y_6, \quad Y_7$$

Part B: Squared-Error Loss

Practice Problem

Problem 4: Squared-Error Loss Calculation

Suppose the predicted outputs are:

$$Y_5 = 1.2, \quad Y_6 = 0.4, \quad Y_7 = 0.8$$

The observed outputs are:

$$Ob_5 = 1, \quad Ob_6 = 0, \quad Ob_7 = 1$$

Use:

$$Loss = \frac{1}{2} [(Ob_5 - Y_5)^2 + (Ob_6 - Y_6)^2 + (Ob_7 - Y_7)^2]$$

Calculate the total loss.

Practice Problem

Problem 5: Output Errors

Given:

$$Ob_5 = 0, \quad Ob_6 = 1, \quad Ob_7 = 0$$

$$Y_5 = 1.5, \quad Y_6 = 0.7, \quad Y_7 = 0.2$$

Calculate:

$$Err_5 = Ob_5 - Y_5$$

$$Err_6 = Ob_6 - Y_6$$

$$Err_7 = Ob_7 - Y_7$$

Then calculate:

$$\frac{\partial Loss}{\partial Y_5}, \quad \frac{\partial Loss}{\partial Y_6}, \quad \frac{\partial Loss}{\partial Y_7}$$

using:

$$\frac{\partial Loss}{\partial Y_j} = -Err_j$$

Part C: Output Layer Backpropagation with ReLU Output

Practice Problem

Problem 6: Output Delta with ReLU

Suppose:

$$Err_5 = -1.5, \quad Err_6 = 0.3, \quad Err_7 = -0.2$$

and:

$$z_5 = 2.0, \quad z_6 = -0.5, \quad z_7 = 0.8$$

For ReLU output, calculate:

$$\delta_5 = Err_5 \text{ReLU}'(z_5)$$

$$\delta_6 = Err_6 \text{ReLU}'(z_6)$$

$$\delta_7 = Err_7 \text{ReLU}'(z_7)$$

Then explain why one of the deltas becomes zero.

Practice Problem

Problem 7: Output Weight Update

Given:

$$Y_3 = 0.8, \quad Y_4 = 0.25$$

$$\delta_5 = -1.3, \quad \eta = 0.1$$

$$W_{35} = -0.1, \quad W_{45} = 1.5, \quad b_5 = 1$$

Use:

$$W_{35}^{new} = W_{35} + \eta Y_3 \delta_5$$

$$W_{45}^{new} = W_{45} + \eta Y_4 \delta_5$$

$$b_5^{new} = b_5 + \eta \delta_5$$

Calculate the updated values of:

$$W_{35}^{new}, \quad W_{45}^{new}, \quad b_5^{new}$$

Practice Problem

Problem 8: All Output Layer Updates

Given:

$$Y_3 = 1.0, \quad Y_4 = 0.5, \quad \eta = 0.2$$

$$\delta_5 = -0.4, \quad \delta_6 = 0.6, \quad \delta_7 = -0.2$$

Initial weights and biases:

$$W_{35} = 0.3, \quad W_{45} = 0.7, \quad b_5 = 0.1$$

$$W_{36} = -0.5, \quad W_{46} = 0.2, \quad b_6 = -0.1$$

$$W_{37} = 0.4, \quad W_{47} = -0.6, \quad b_7 = 0.2$$

Calculate all updated output-layer parameters:

$$W_{35}, W_{45}, b_5$$

$$W_{36}, W_{46}, b_6$$

$$W_{37}, W_{47}, b_7$$

Part D: Hidden Layer Backpropagation

Practice Problem

Problem 9: Hidden Delta for Node 3

Given:

$$\delta_5 = -1.0, \quad \delta_6 = 0.5, \quad \delta_7 = -0.2$$

$$W_{35} = 0.4, \quad W_{36} = -0.3, \quad W_{37} = 0.2$$

$$z_3 = 0.7$$

Use:

$$\delta_3 = (\delta_5 W_{35} + \delta_6 W_{36} + \delta_7 W_{37}) \text{ReLU}'(z_3)$$

Calculate δ_3 .

Practice Problem

Problem 10: Hidden Delta When ReLU Blocks Gradient

Given:

$$\delta_5 = 0.8, \quad \delta_6 = -0.4, \quad \delta_7 = 0.3$$

$$W_{45} = 0.5, \quad W_{46} = 0.1, \quad W_{47} = -0.2$$

$$z_4 = -0.6$$

Use:

$$\delta_4 = (\delta_5 W_{45} + \delta_6 W_{46} + \delta_7 W_{47}) \text{ReLU}'(z_4)$$

Calculate δ_4 . Explain why the answer becomes zero.

Practice Problem

Problem 11: Hidden Layer Weight Update

Given:

$$X_1 = 0.5, \quad X_2 = 0.75, \quad \eta = 0.1$$

$$\delta_3 = 0.25$$

$$W_{13} = -2.5, \quad W_{23} = 0.6, \quad b_3 = 1.6$$

Use:

$$W_{13}^{new} = W_{13} + \eta X_1 \delta_3$$

$$W_{23}^{new} = W_{23} + \eta X_2 \delta_3$$

$$b_3^{new} = b_3 + \eta \delta_3$$

Calculate:

$$W_{13}^{new}, \quad W_{23}^{new}, \quad b_3^{new}$$

Part E: Complete ReLU-Output Backpropagation Problem

Practice Problem

Problem 12: Complete One-Step Update with ReLU Output

Given:

$$X_1 = 0.5, \quad X_2 = 0.75$$

Learning rate:

$$\eta = 0.1$$

Hidden layer:

$$W_{13} = -2.5, \quad W_{23} = 0.6, \quad b_3 = 1.6$$

$$W_{14} = -1.5, \quad W_{24} = 0.4, \quad b_4 = 0.7$$

Output layer:

$$W_{35} = -0.1, \quad W_{45} = 1.5, \quad b_5 = 1$$

$$W_{36} = 2.4, \quad W_{46} = -5.2, \quad b_6 = 0.5$$

$$W_{37} = -2.2, \quad W_{47} = 3.7, \quad b_7 = 1$$

Observed output:

$$Ob_5 = 0, \quad Ob_6 = 1, \quad Ob_7 = 0$$

The hidden layer and output layer both use ReLU.

Answer the following:

- (a) Compute z_3, z_4, Y_3, Y_4 .
- (b) Compute $z_5, z_6, z_7, Y_5, Y_6, Y_7$.
- (c) Compute Err_5, Err_6, Err_7 .
- (d) Compute $\delta_5, \delta_6, \delta_7$.
- (e) Update W_{35}, W_{45}, b_5 .
- (f) Compute δ_3, δ_4 .
- (g) Update W_{13}, W_{23}, b_3 .
- (h) Update W_{14}, W_{24}, b_4 .

Part F: Softmax and Cross-Entropy

Practice Problem

Problem 13: Softmax Probability Calculation

Given logits:

$$z_5 = 2, \quad z_6 = 1, \quad z_7 = 0$$

Calculate:

$$P_5 = \frac{e^{z_5}}{e^{z_5} + e^{z_6} + e^{z_7}}$$

$$P_6 = \frac{e^{z_6}}{e^{z_5} + e^{z_6} + e^{z_7}}$$

$$P_7 = \frac{e^{z_7}}{e^{z_5} + e^{z_6} + e^{z_7}}$$

Then state the predicted class.

Practice Problem

Problem 14: Cross-Entropy Loss

Suppose the correct class is class 5.

Given:

$$P_5 = 0.70, \quad P_6 = 0.20, \quad P_7 = 0.10$$

Use:

$$Loss = -\log(P_5)$$

Calculate the loss.

Then explain what would happen to the loss if P_5 became closer to 1.

Practice Problem

Problem 15: Softmax Gradient

Suppose the correct class is class 5.

Target vector:

$$t_5 = 1, \quad t_6 = 0, \quad t_7 = 0$$

Given:

$$P_5 = 0.6, \quad P_6 = 0.3, \quad P_7 = 0.1$$

Use:

$$\frac{\partial \text{Loss}}{\partial z_j} = P_j - t_j$$

Calculate:

$$\frac{\partial \text{Loss}}{\partial z_5}$$

$$\frac{\partial \text{Loss}}{\partial z_6}$$

$$\frac{\partial \text{Loss}}{\partial z_7}$$

Practice Problem

Problem 16: Softmax Output Layer Update

Given:

$$Y_3 = 0.8, \quad Y_4 = 0.25, \quad \eta = 0.1$$

Softmax probabilities:

$$P_5 = 0.6, \quad P_6 = 0.3, \quad P_7 = 0.1$$

Correct class:

Class 5

Initial output weights and biases:

$$W_{35} = -0.1, \quad W_{45} = 1.5, \quad b_5 = 1$$

$$W_{36} = 2.4, \quad W_{46} = -5.2, \quad b_6 = 0.5$$

$$W_{37} = -2.2, \quad W_{47} = 3.7, \quad b_7 = 1$$

Use:

$$W_{35}^{new} = W_{35} - \eta Y_3 (P_5 - 1)$$

$$W_{45}^{new} = W_{45} - \eta Y_4 (P_5 - 1)$$

$$b_5^{new} = b_5 - \eta (P_5 - 1)$$

$$W_{36}^{new} = W_{36} - \eta Y_3 P_6$$

$$W_{46}^{new} = W_{46} - \eta Y_4 P_6$$

$$b_6^{new} = b_6 - \eta P_6$$

$$W_{37}^{new} = W_{37} - \eta Y_3 P_7$$

$$W_{47}^{new} = W_{47} - \eta Y_4 P_7$$

$$b_7^{new} = b_7 - \eta P_7$$

Calculate all updated output-layer parameters.

Part G: Softmax Hidden Layer Backpropagation

Practice Problem

Problem 17: Hidden Delta for Softmax Output

Suppose the correct class is class 5.

Given:

$$P_5 = 0.6, \quad P_6 = 0.3, \quad P_7 = 0.1$$

$$W_{35} = -0.1, \quad W_{36} = 2.4, \quad W_{37} = -2.2$$

$$z_3 = 0.8$$

Use:

$$\delta_3 = [(P_5 - 1)W_{35} + P_6W_{36} + P_7W_{37}] \text{ReLU}'(z_3)$$

Calculate δ_3 .

Practice Problem

Problem 18: Hidden Delta for Node 4 in Softmax Case

Suppose the correct class is class 5.

Given:

$$P_5 = 0.6, \quad P_6 = 0.3, \quad P_7 = 0.1$$

$$W_{45} = 1.5, \quad W_{46} = -5.2, \quad W_{47} = 3.7$$

$$z_4 = 0.25$$

Use:

$$\delta_4 = [(P_5 - 1)W_{45} + P_6W_{46} + P_7W_{47}] \text{ReLU}'(z_4)$$

Calculate δ_4 .

Practice Problem

Problem 19: Softmax Hidden Weight Update

Given:

$$X_1 = 0.5, \quad X_2 = 0.75, \quad \eta = 0.1$$

$$\delta_3 = 0.54$$

Initial hidden parameters:

$$W_{13} = -2.5, \quad W_{23} = 0.6, \quad b_3 = 1.6$$

Use:

$$W_{13}^{new} = W_{13} - \eta \delta_3 X_1$$

$$W_{23}^{new} = W_{23} - \eta \delta_3 X_2$$

$$b_3^{new} = b_3 - \eta \delta_3$$

Calculate:

$$W_{13}^{new}, \quad W_{23}^{new}, \quad b_3^{new}$$

Part H: Conceptual Questions

Practice Problem

Problem 20: Conceptual

Answer briefly.

- (a) What is the difference between forward pass and backward pass?
- (b) Why do we need an activation function such as ReLU?
- (c) What happens to the gradient when $z < 0$ for ReLU?
- (d) Why does a hidden node receive error from multiple output nodes?
- (e) Why is softmax useful for multi-class classification?
- (f) What does cross-entropy loss encourage the model to do?
- (g) Why do we update parameters in the opposite direction of the gradient?

Part I: Problem

Practice Problem

Problem 21: One Complete MLP Problem

A neural network has two inputs, two hidden nodes, and three output nodes.

$$X_1 = 1, \quad X_2 = 2$$

Hidden layer:

$$W_{13} = 0.4, \quad W_{23} = 0.1, \quad b_3 = -0.2$$

$$W_{14} = -0.3, \quad W_{24} = 0.5, \quad b_4 = 0.1$$

Output logits:

$$W_{35} = 0.2, \quad W_{45} = 0.4, \quad b_5 = 0.1$$

$$W_{36} = -0.5, \quad W_{46} = 0.3, \quad b_6 = -0.2$$

$$W_{37} = 0.6, \quad W_{47} = -0.1, \quad b_7 = 0.0$$

The hidden layer uses ReLU. The output layer uses softmax.

The correct class is class 7.

Learning rate:

$$\eta = 0.1$$

Answer the following:

- (a) Compute z_3, z_4, Y_3, Y_4 .
- (b) Compute output logits z_5, z_6, z_7 .
- (c) Compute softmax probabilities P_5, P_6, P_7 .
- (d) Compute cross-entropy loss.
- (e) Compute output gradients:

$$\frac{\partial \text{Loss}}{\partial z_5}, \frac{\partial \text{Loss}}{\partial z_6}, \frac{\partial \text{Loss}}{\partial z_7}$$

- (f) Update W_{37}, W_{47}, b_7 .
- (g) Compute hidden delta δ_3 .
- (h) Update W_{13}, W_{23}, b_3 .